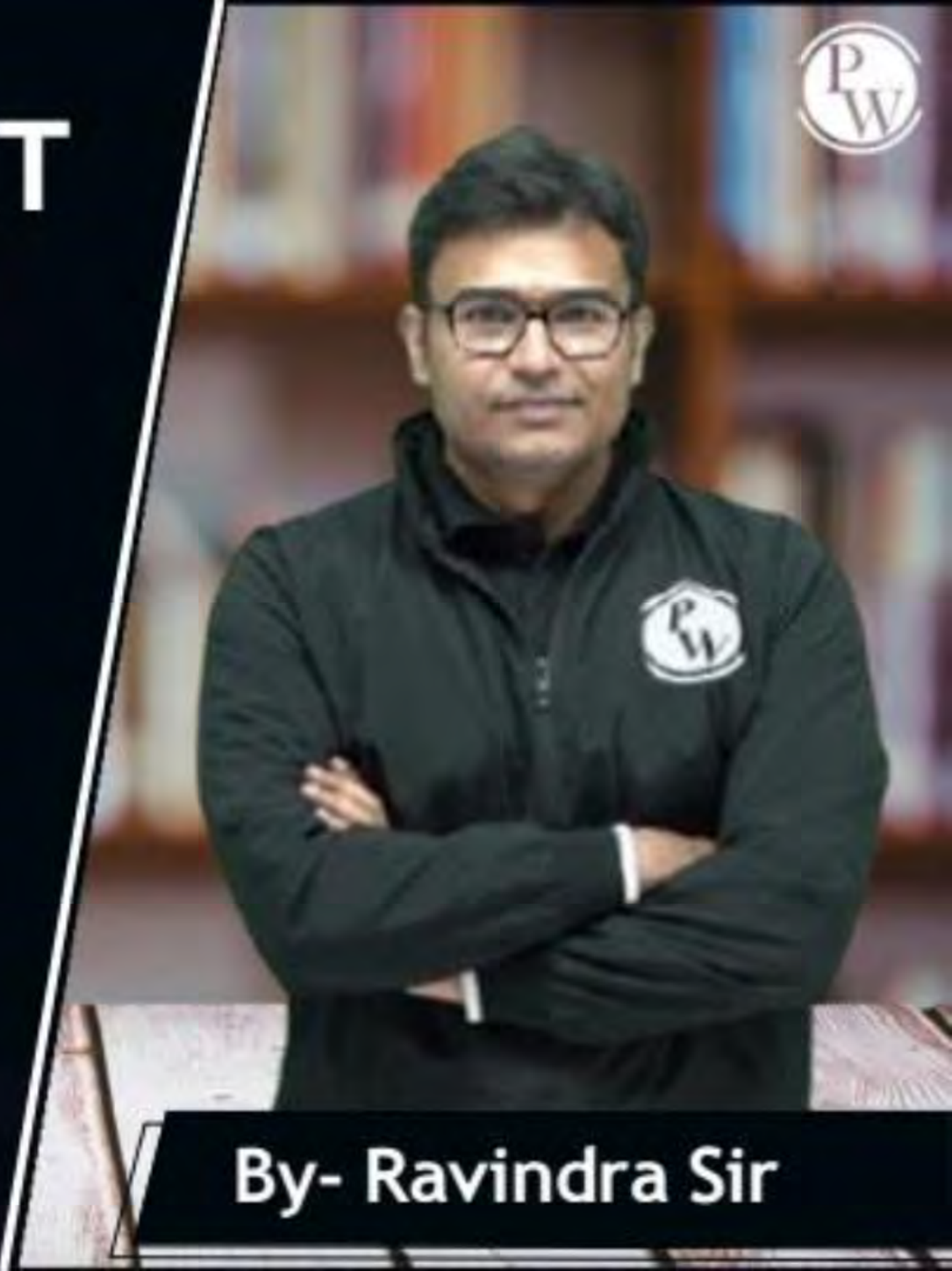
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 06



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

Mag

Topics to be Covered



- Topic
- Topic
- Topic

Fragmentation

Inspiring Stories : Chen Si



Background: Factory worker from Nanjing, China..

Education: Basic schooling.

Achievements: Spends weekends at a bridge stopping suicides, offering help.

Impact: Saved hundreds of lives through quiet kindness.

Inspiring Stories : Betty Makoni



Background: Grew up in hardship in Zimbabwe.

Education: Studied while working.

Achievements: Founded Girl Child Network to protect unsafe girls.

Impact: Gave safety and voice to thousands who had none.

Inspiring Stories : P. Bai Yadav



Background: Born in a poor family in Chhattisgarh.

Education: Dropped out early.

Achievements: Formed women's groups to fight poverty and crime; runs self-help programs.

Impact: Improved thousands of women's lives in rural India.

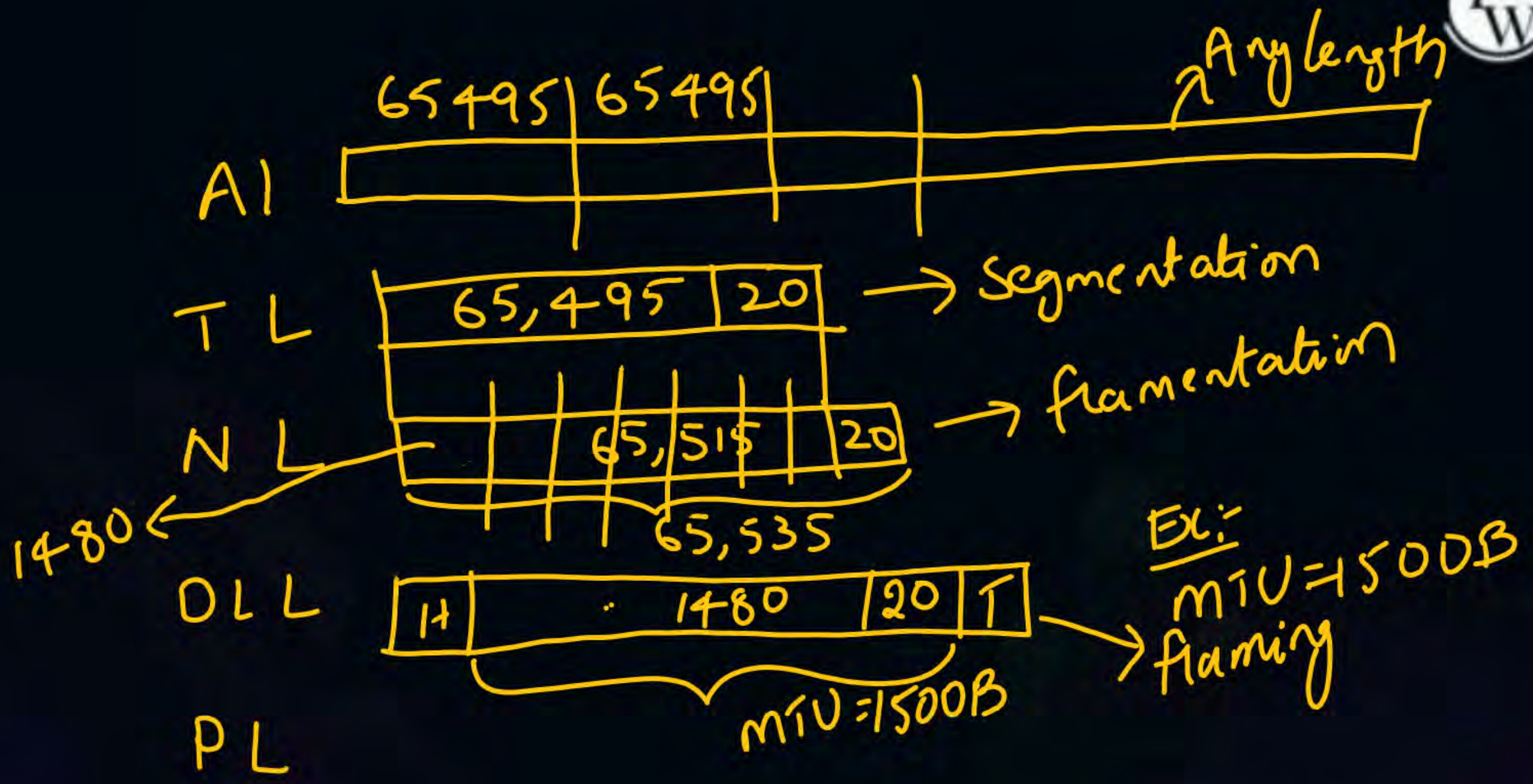
Ipv4 : Total length = 16 bits

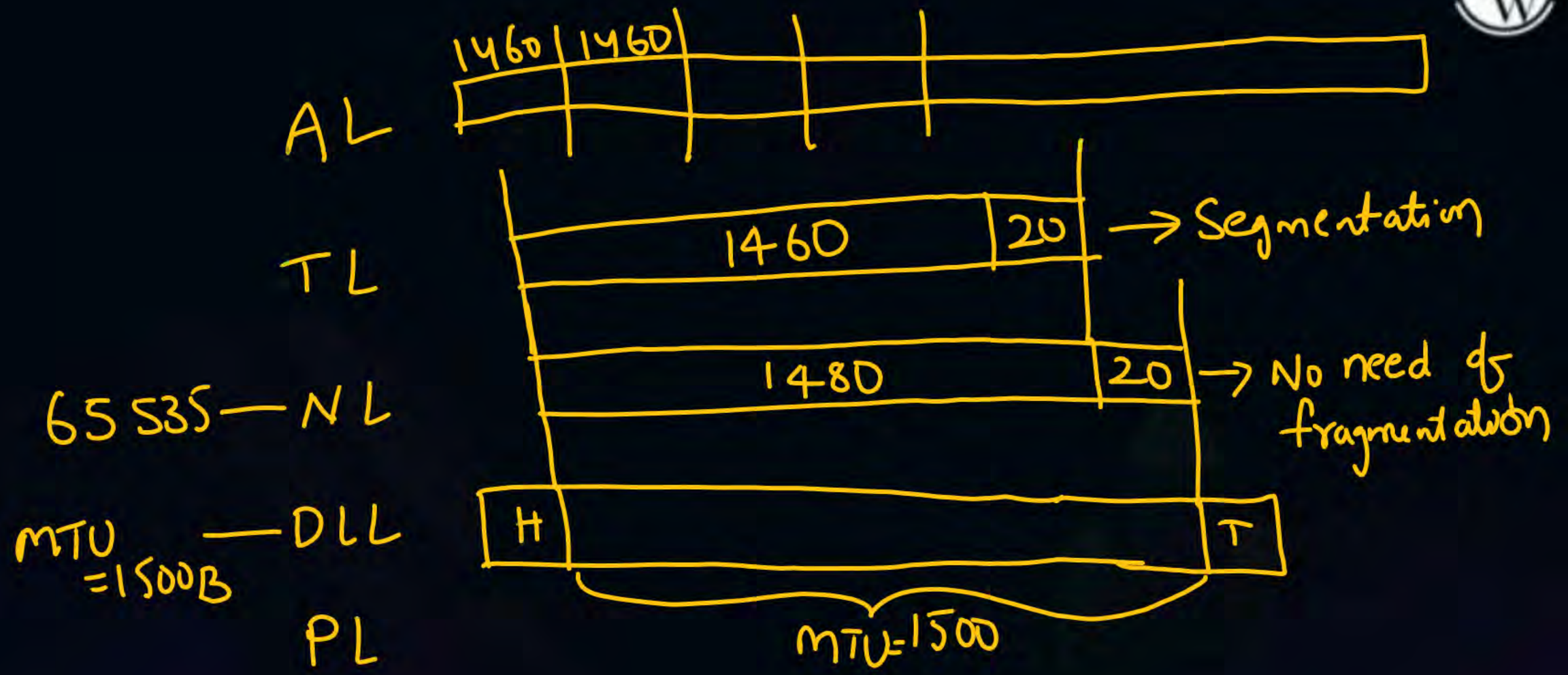
max number = $2^{16} - 1$

$= 65535$

↓

maximum length of a IP datagram

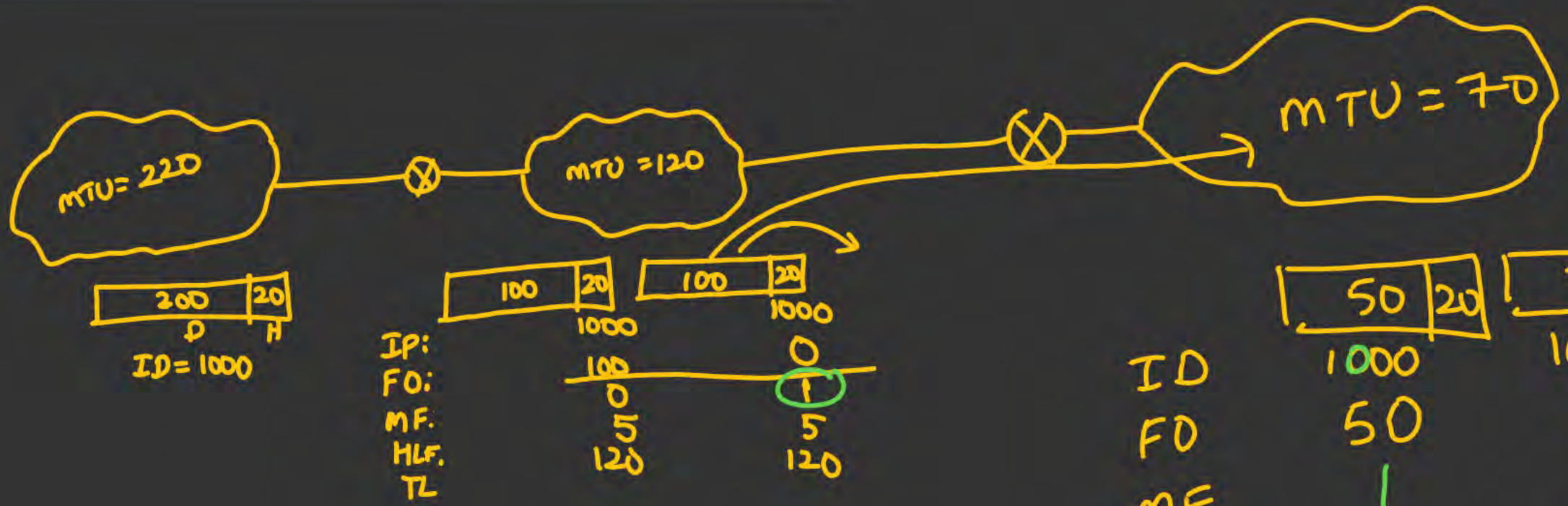

65495
65495
Any length
AI
TL
65,495
20
Segmentation
NL
65,515
20
Flamentation
1480
65,535
OLL
IH
1480
20
T
Ex:-
MTU=1500B
Flaming
PL
MTU=1500B


AL
1460
1460
TL
1460
20
→ Segmentation
65535 — NL
1480
20
→ No need of fragmentation
MTU =1500B
— DLL
H
T
PL
MTU=1500

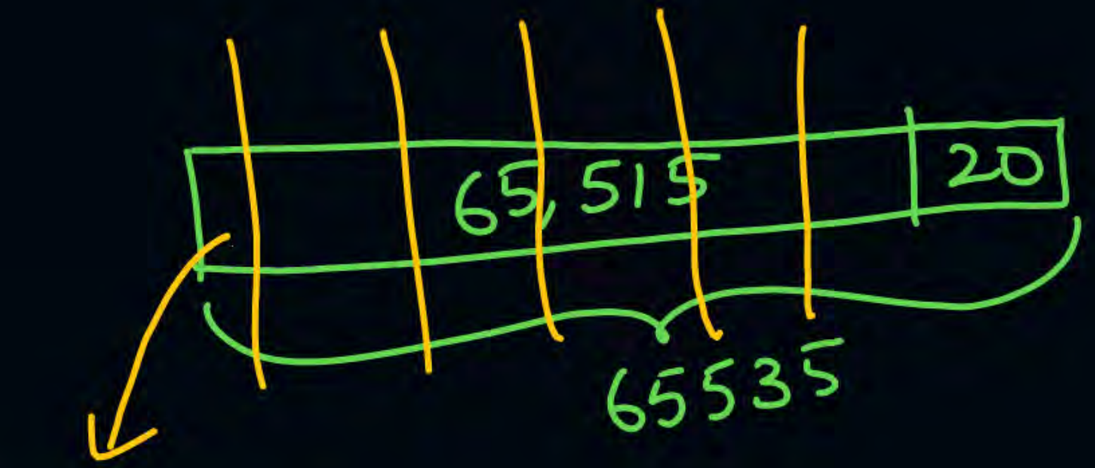
Source will not do fragmentation if Segmentation is done wisely

Q) Where is fragmentation done?

A) Routers



ID	1000	1000
FO	50	0
MF	1	1
HLF	5	5
TL	70	70





FO: 65,514 → worst case number of FO

$\simeq 2^{16}$

FO → 13 bits 2^{13}

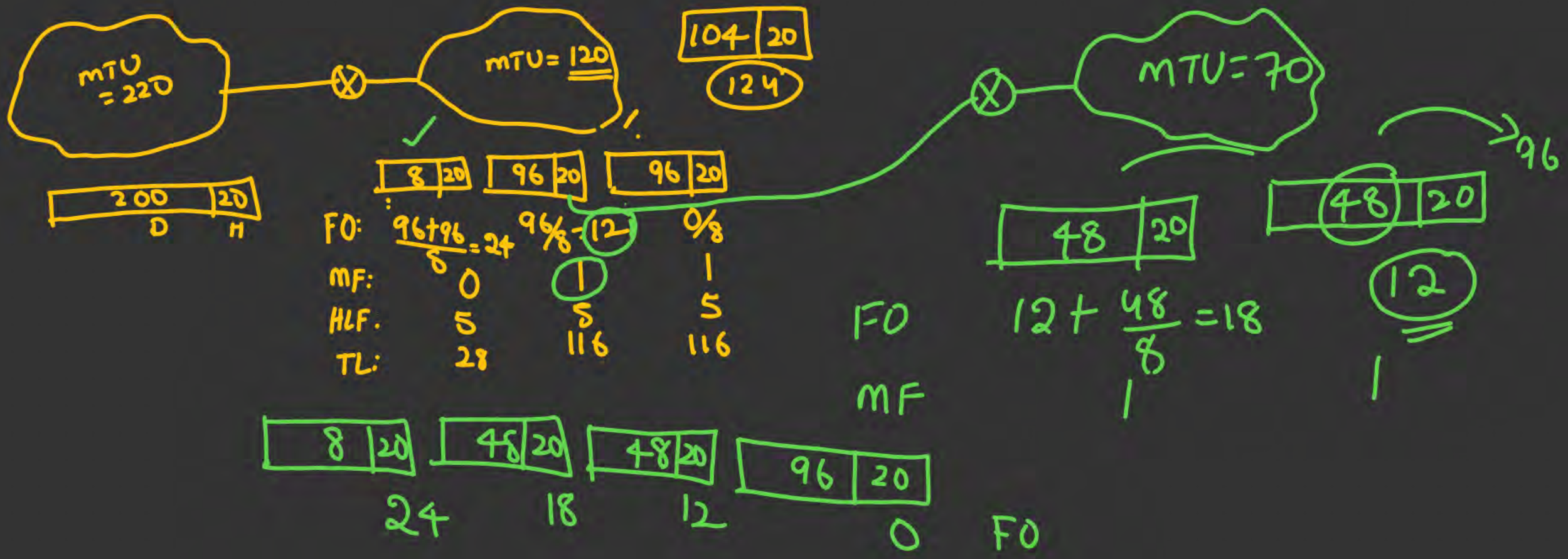
$\frac{2^{16}}{2^{13}} = 8$ → scaling factor

The maximum number that FO should represent is $65,514 \cong 2^{16}$

But FO has only 13 bits $\longrightarrow 2^{13}-1$

$$\frac{2^{16}}{2^{13}} = (8) \longrightarrow \text{Scaling factor}$$

The actual FO is divided by 8, and stored in the Fragment offset field.


MTU = 220
MTU = 120
MTU = 70
200 20
D H
104 20
124
8 20
96 20
96 20
FO: 96+96/8 = 24
96/8 = 12
0/8
MF: 0
1
1
HLF. 5
5
5
TL: 28
116
116
48 20
12 + 48/8 = 18
48 20
12
96
1
1
FO
MF
8 20
48 20
48 20
96 20
24
18
12
0
FO

→ Every fragment should contain header

Q) How will destination identify that all the fragments belong to the same datagram?

A) Identification number (16 bits) $\cong 2^{16}$ Identification numbers are possible

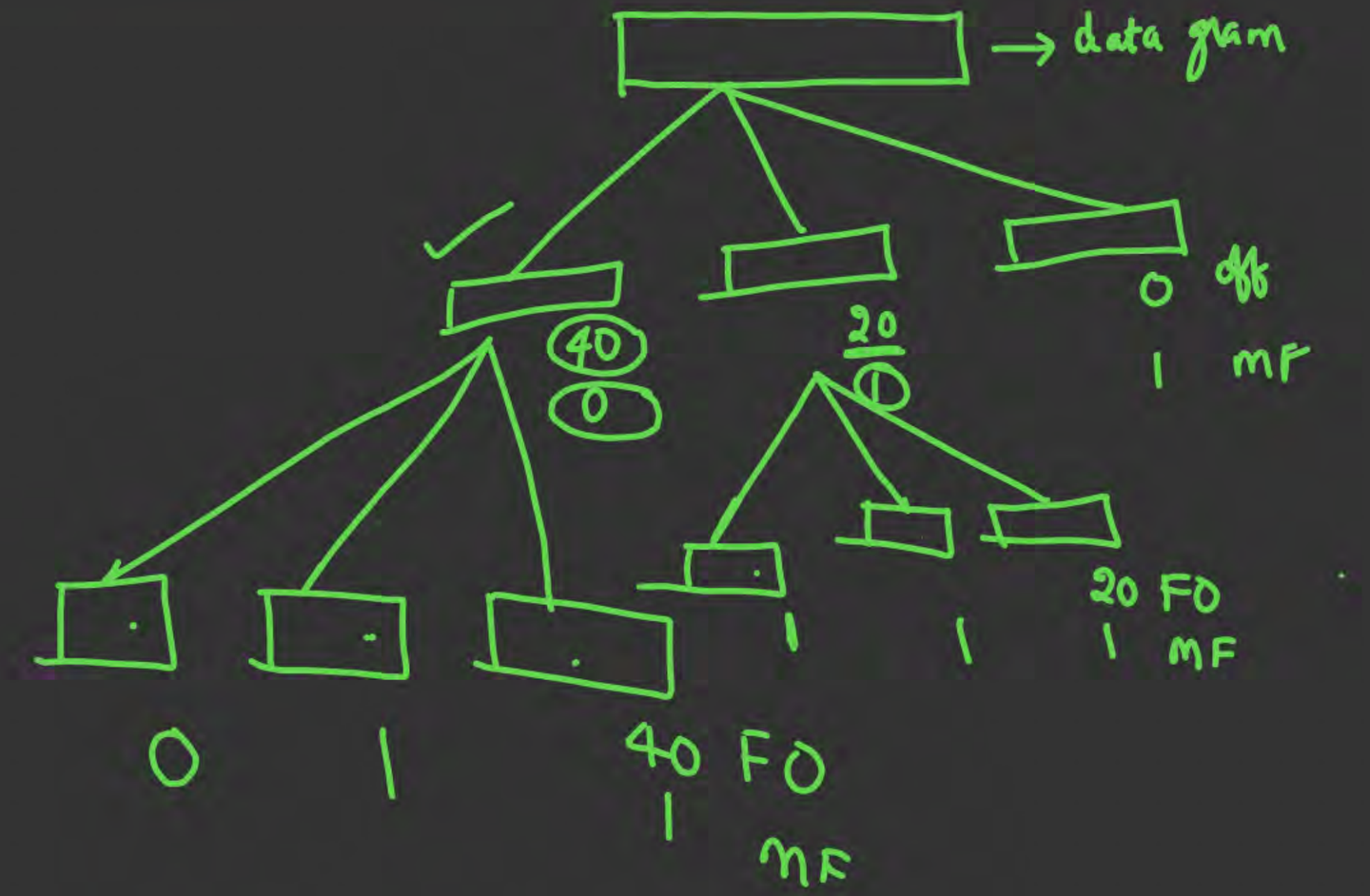
Q) How will destination identify the sequence of flaments?

A) Fragment offset

Fragment offset of a fragment is no of (data bytes ahead of it)/8

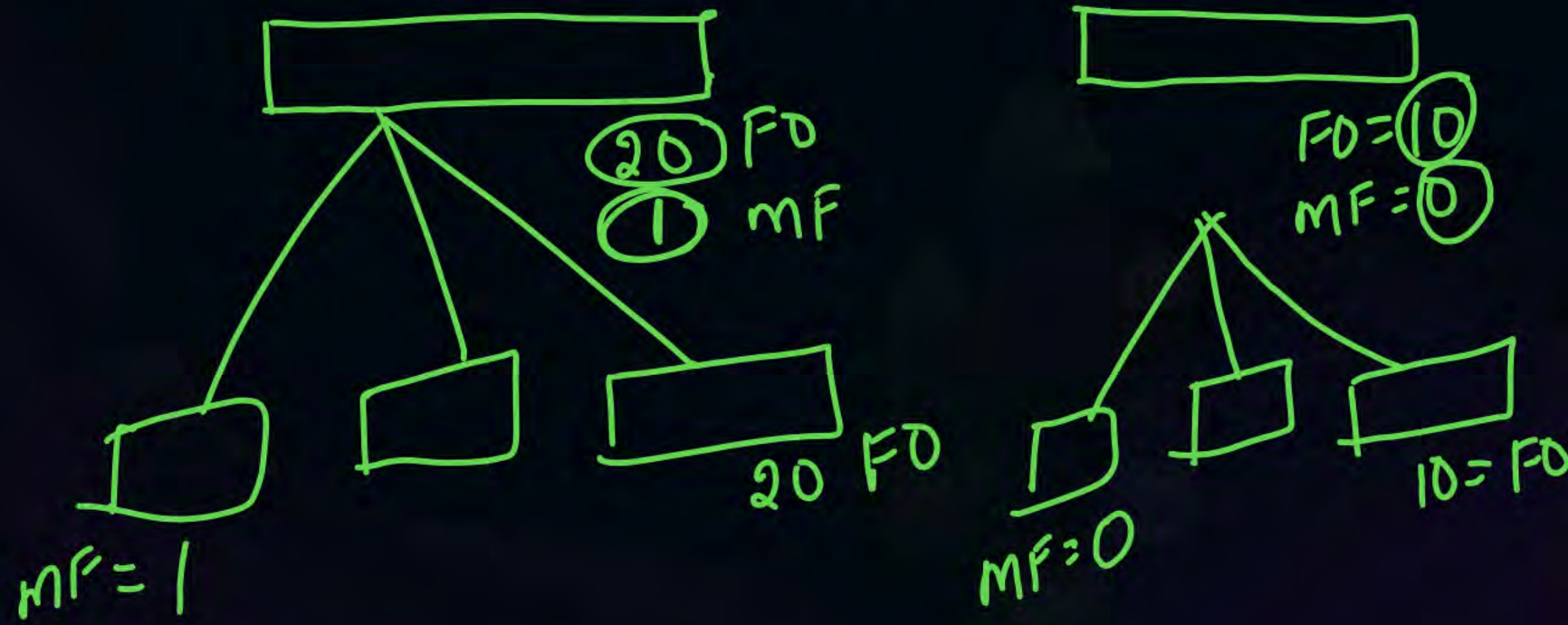
→ How can we identify how many data bytes are present in a fragment of datagram

A) Total length − header length

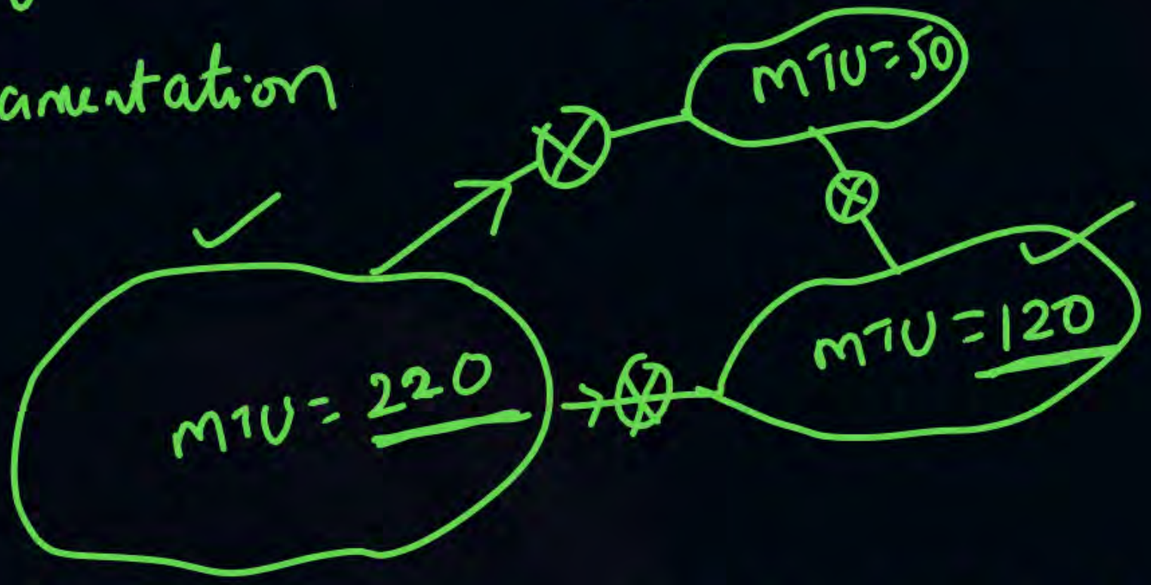

data gram
40
0
20
1
0
off
1
MF
0
1
40
FO
1
MF
20
FO
1
MF

→ when a fragment is fragmented, the first fragment will get same FO as the parent fragment

→ when a fragment is fragmented, MF of the last fragment will be same as MF of the parent fragment

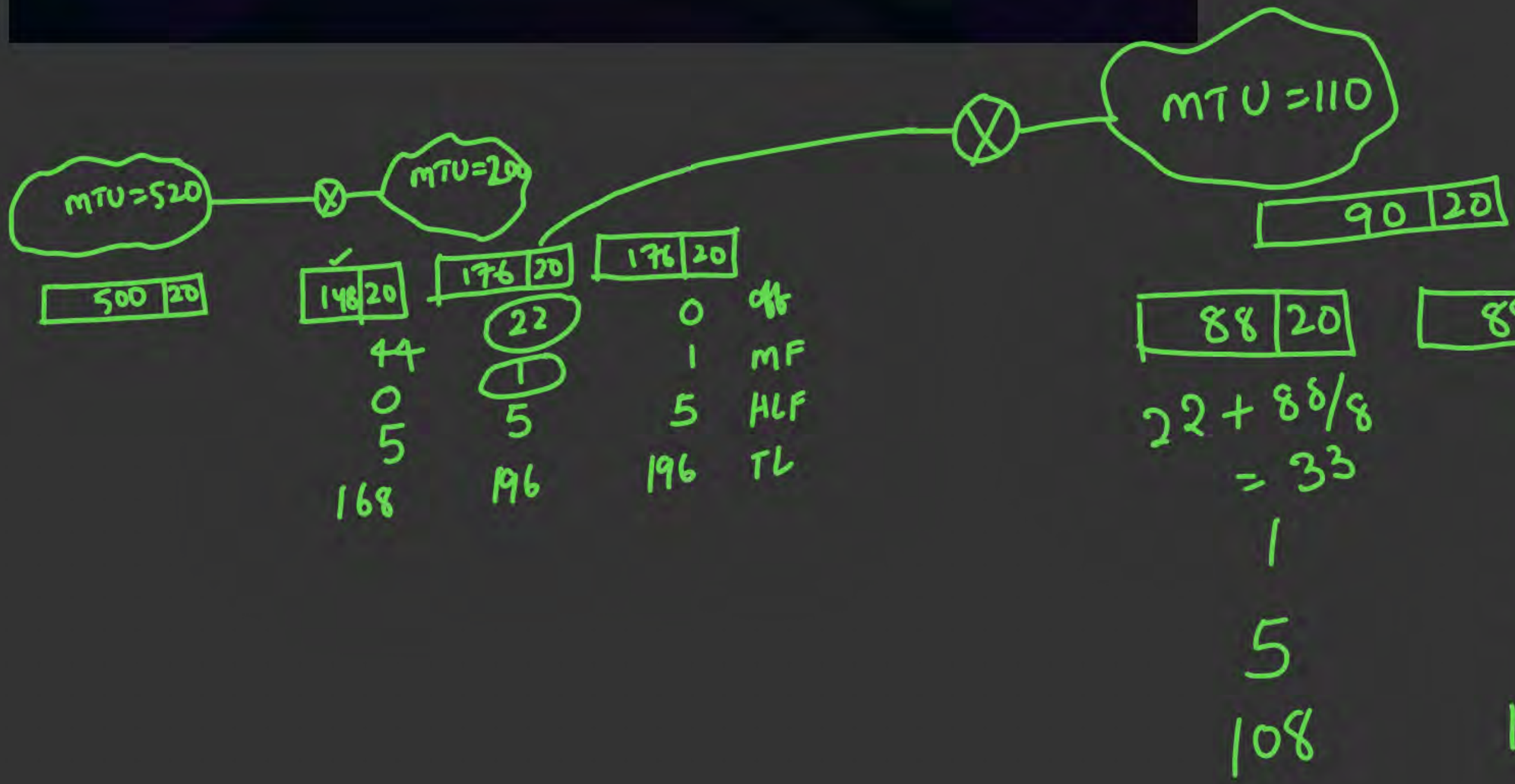


→ Why Can't we do fragmentation such that there is no need of fragmentation



A source will know MTU of only its network.

Also the route may change.


MTU=520
MTU=200
MTU=110
500 20
148 20
176 20
176 20
90 20
88 20
88 20
44
0
5
168
22
1
5
196
0
1
5
196
off
MF
HLF
TL
22 + 88/8
= 33
1
5
108
22
1
5
108
off
MF
HLF
TL

Last fragment size need not be multiple of 8.



$$\eta = \frac{\text{Useful bytes}}{\text{Total bytes}} = \frac{500}{580}$$

Off MF

BW util
Eff BW
Thruput
$\eta * BW$

Q) Where is fragmentation done?

A) Routers

Q) Where are fragments reassembled?

A) Destination

Q) Why is reassembling not done at the routers?

A) All the fragments may not meet at a router

Even if reassembly is done at router, there may be fragmentation required again.





20 = Off

1 = MF

5 = HL

108 = TL

8	20

40	20

40	20

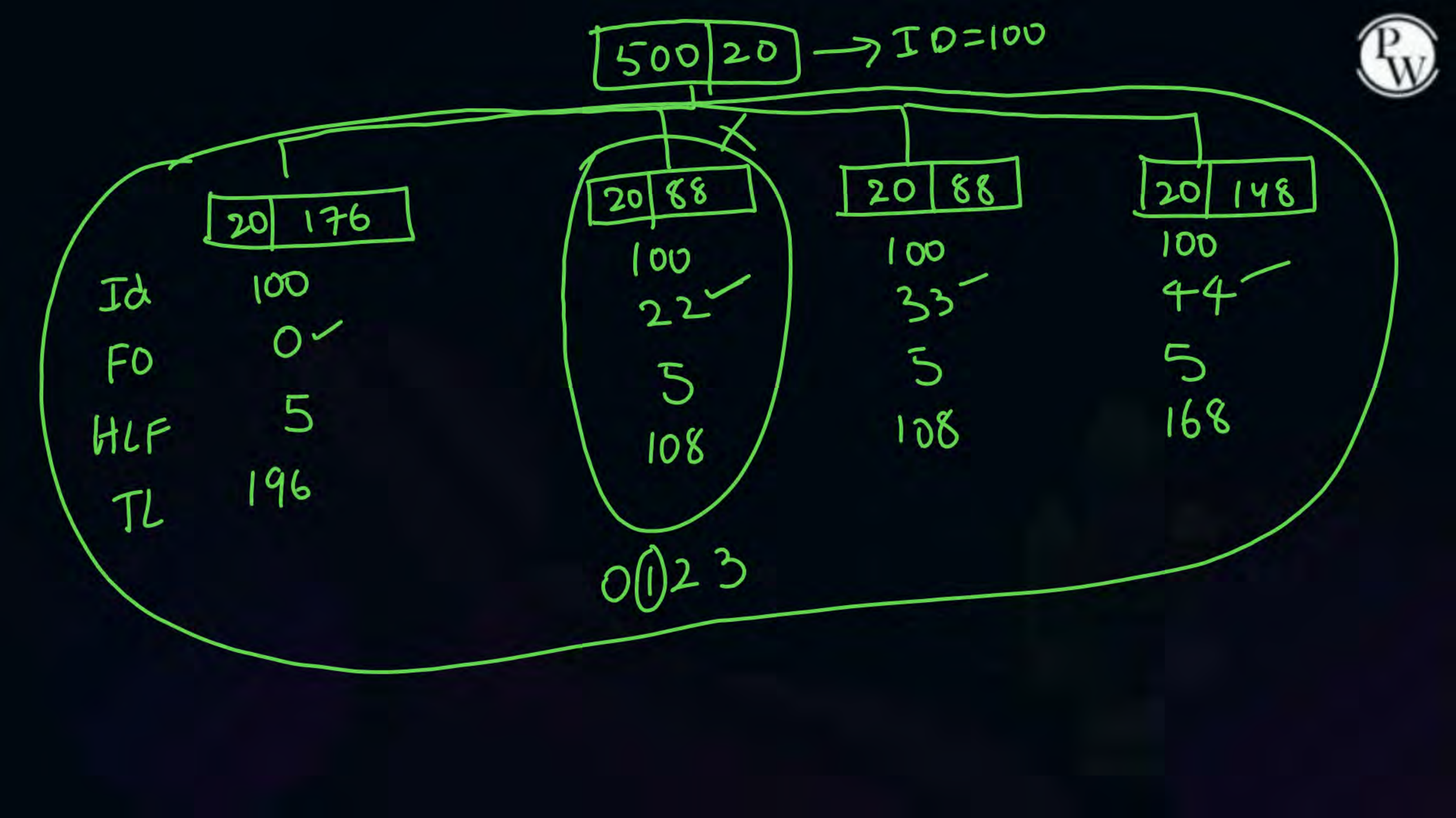
25 + 40/8 = 30

20 + 40/8 = 25

1	1	1	MF
5	5	5	HL
28	60	60	TL

20

Off


500 | 20
ID=100
20 | 176
20 | 88
20 | 88
20 | 148
Id
FO
HLF
TL
100
0
5
196
100
22
5
108
100
33
5
108
100
44
5
168
0 1 2 3

THANK - YOU